Introduction
PART 1

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What is Histology?

- The science that study the structure of the cells & tissues using Microscope

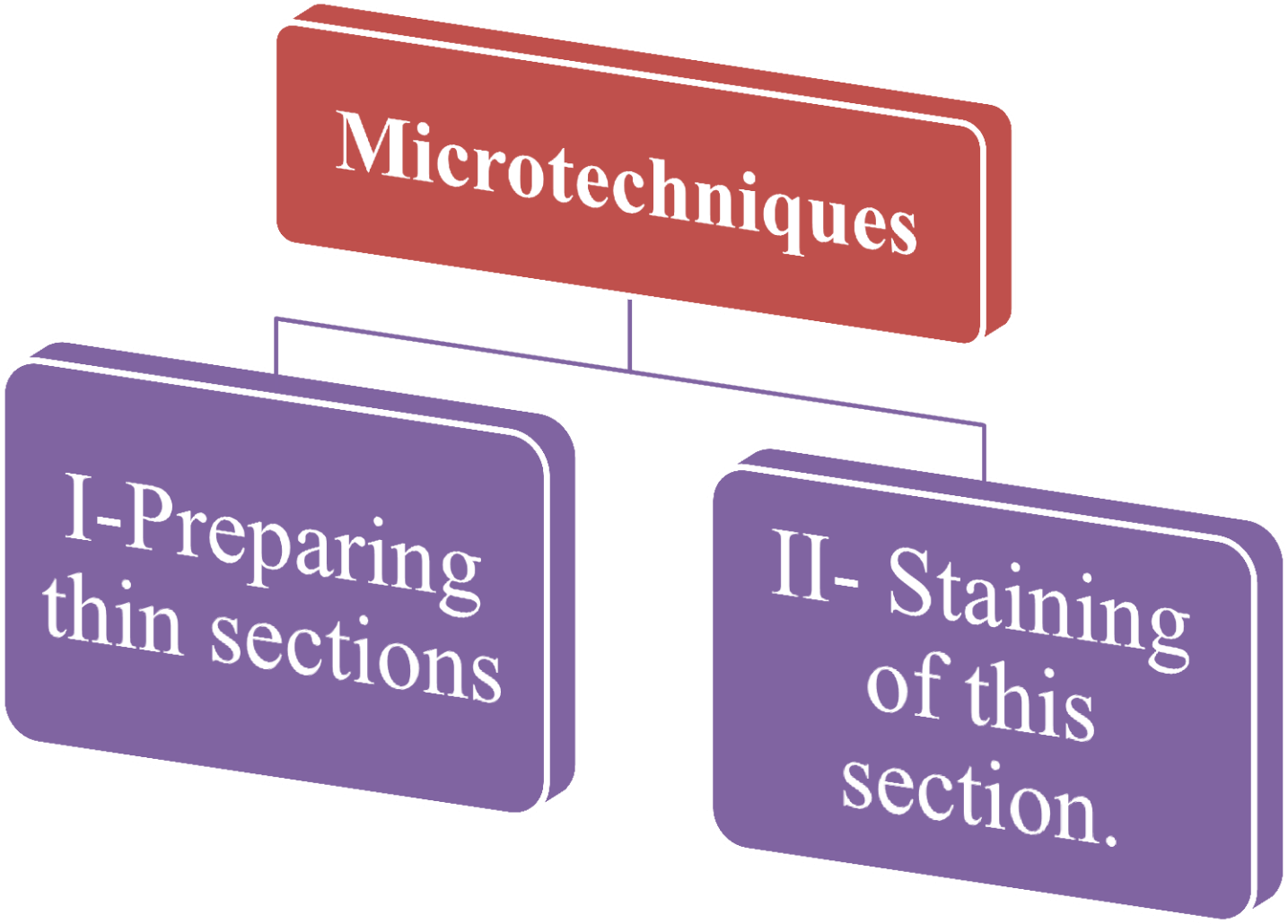


Why do we study it?

- Study microscopic structure of cells & tissues
- Correlate this structure to function
- Understand the pathology of diseases

Microtechniques.

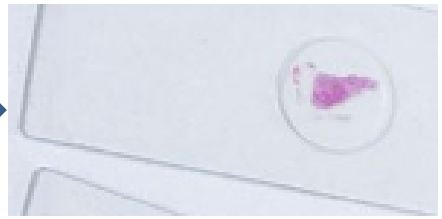



Microtechniques
I-Preparing thin sections
II- Staining of this section.

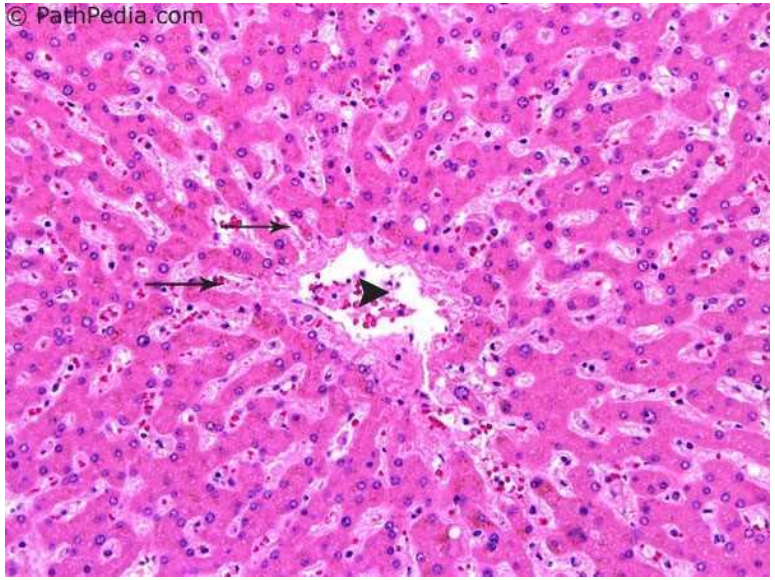
Microtechniques











I. Preparing thin section

Paraffin technique.

- The routine method for histological preparation.
- Paraffin wax is used.
- **Steps:**

1- Obtain sample



2- Fixation.

- Put small sample of tissue in fixative e.g. Formalin 10%.

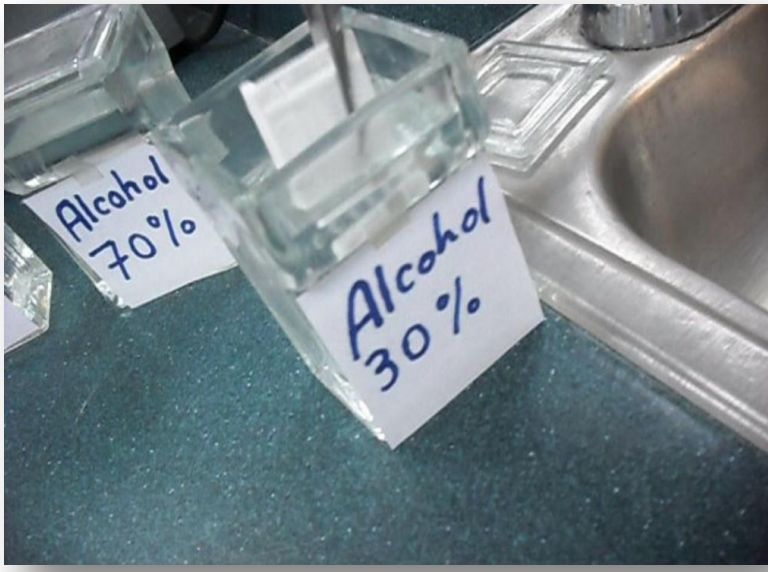
➢Prevents putrefaction and autolysis.



3-Dehydration.

- In ascending grades of alcohol (70%-90%-100% alcohol).

➢Avoid shrinkage of the tissue.





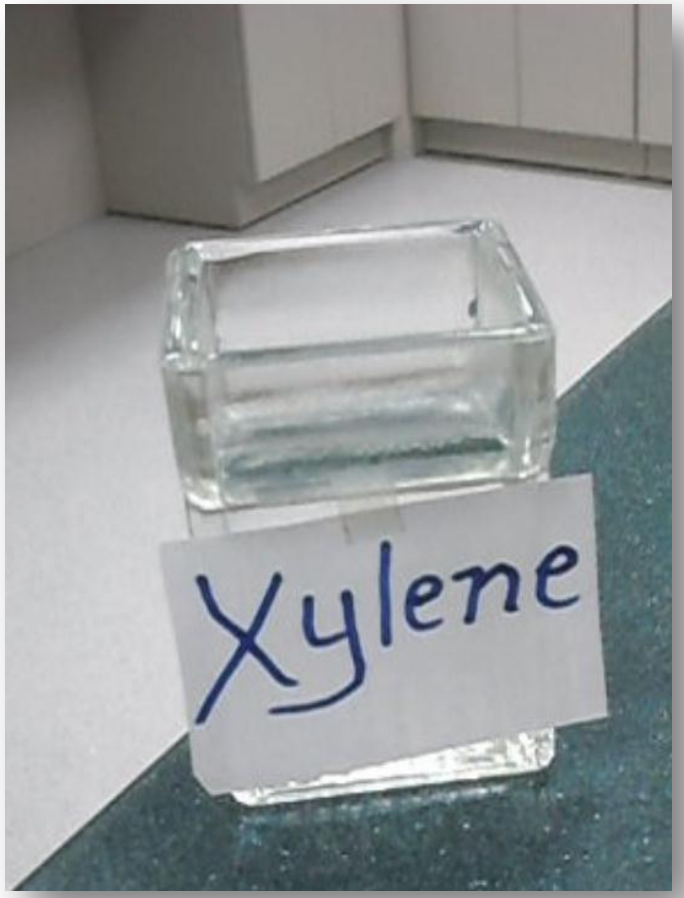
4-Clearing.

- With Xylol to:

➢Replace alcohol

➢Paraffin solvent

➢Renders the tissue transparent.



5-Impregnation.

- Infiltration with Paraffin wax in the oven.





6-Embedding in Paraffin

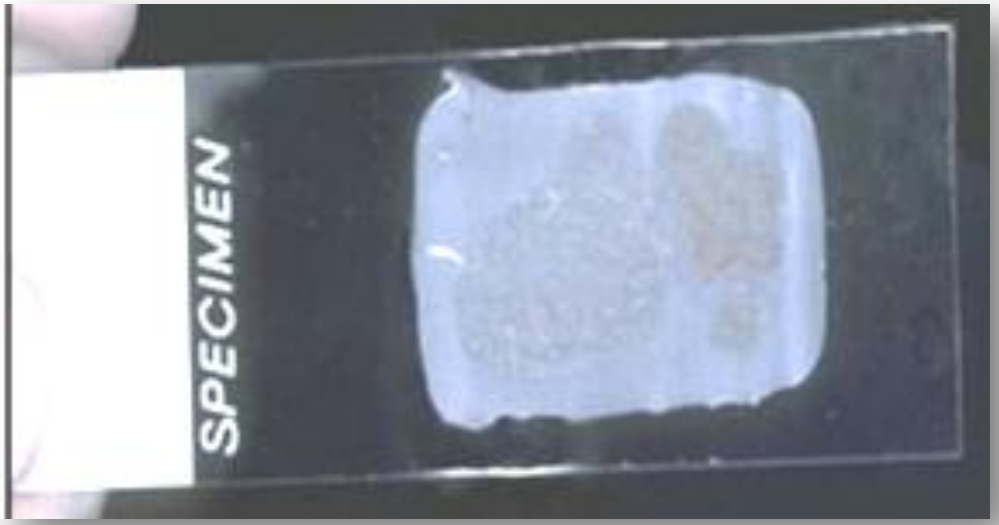
- In hard paraffin to obtain a paraffin block.



7- Sectioning .

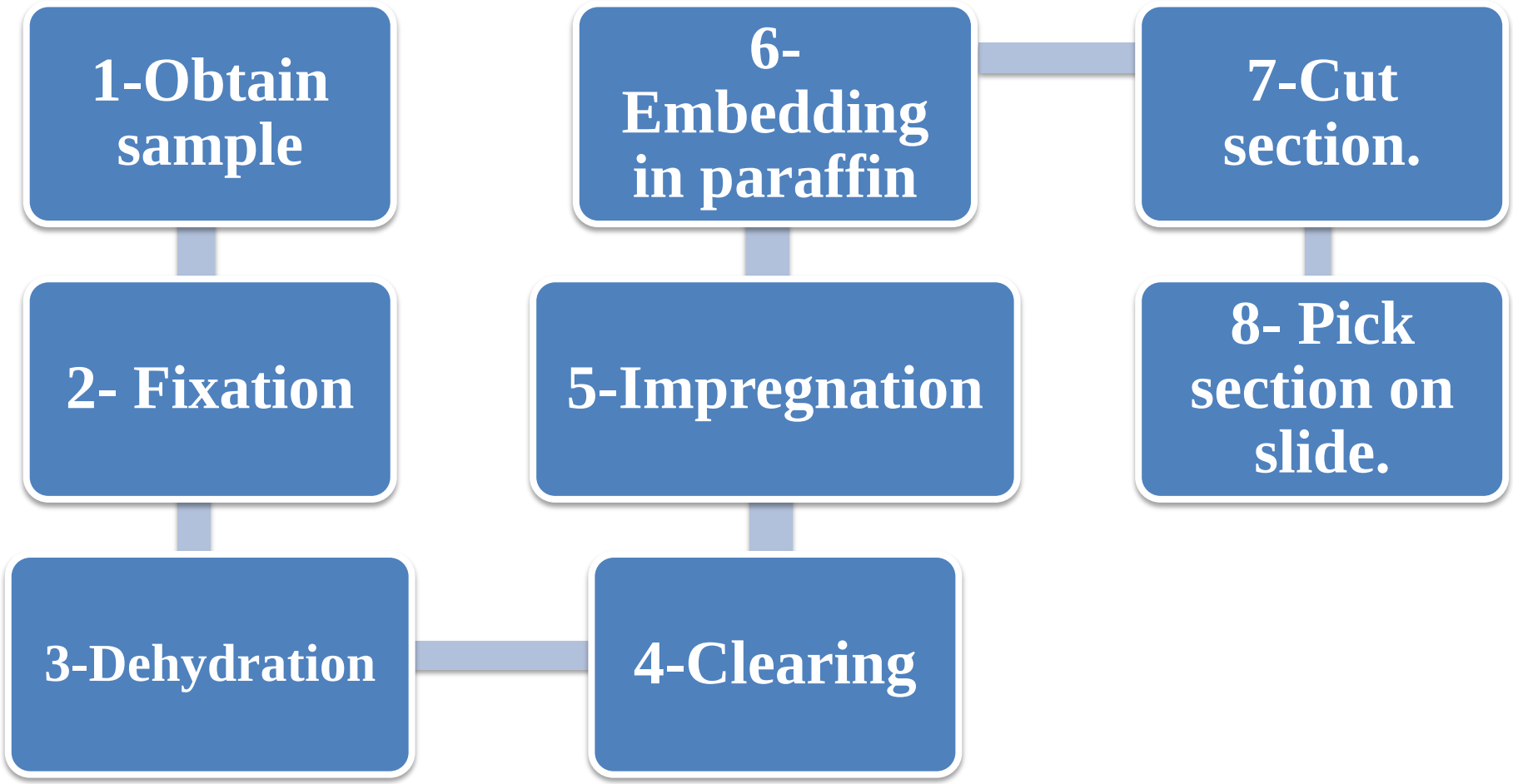






Thickness of section is 5 um

Paraffin technique.




Thank
You